

HUM 4201: PRINCIPLES OF ECONOMICS

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Chapter-04: Money Growth and Inflation

INTRODUCTION

Measuring Inflation

- In the **1930s**, a small ice-cream cone cost **3 cents**, while a large cone cost **5 cents**.
- Today, buying an ice-cream cone requires **a few dollars**, illustrating how prices increase over time.
- The **persistent rise in the overall price level** of goods and services is called **inflation**.
- Inflation is commonly measured using:
 - **Consumer Price Index (CPI)**
 - **GDP Deflator**
 - Other general price indexes
- In the **United States**, the average annual inflation rate over the past **80 years** has been approximately **3.6%**.
- Although 3.6% appears small each year, over many decades it resulted in about a **17-fold increase** in the overall price level.
- **Key Takeaway:**
 - ✓ Even moderate annual inflation can dramatically increase prices over time due to compounding.

INTRODUCTION

Quantity Theory of Money

- ❑ Explains **why inflation occurs** and **what determines its magnitude**.
- ❑ One of the **Ten Principles of Economics**:
 - **"Prices rise when the government prints too much money."**
- ❑ Supported by leading economists:
 - David Hume
 - Milton Friedman
- ❑ Explains both **moderate inflation** and **hyperinflation**.

Why Is Inflation a Problem?

- Inflation is generally unpopular and was a major public concern in the **United States during the 1970s**.
- In **1974**, Gerald Ford called inflation "**public enemy number one**" and promoted the **WIN (Whip Inflation Now)** campaign.
- However, the **true economic costs of inflation** are more complex than they first appear.

Key Message:

- The chapter explains the causes of inflation through the Quantity Theory of Money and examines whether inflation is always as economically harmful as it is popularly perceived.

THE CLASSICAL THEORY OF INFLATION

- ☑ The **Quantity Theory of Money** is the foundation of the classical theory of inflation.
- ☑ It is called "**classical**" because it was developed by some of the earliest economists.
- ☑ The theory emphasizes that **changes in the money supply are the primary cause of long-run inflation.**
- ☑ **Key Idea:**
 - ✓ Inflation is fundamentally a **monetary phenomenon.**

The Level of Prices and the Value of Money

- Suppose the price of an ice-cream cone increases from **\$0.05 to \$1.00.**
- The increase is **not necessarily because people value ice cream more.**
- A more likely explanation is that **money has become less valuable.**
- **Important Insight:** Inflation is **more about the declining value of money** than the increasing value of goods.

THE CLASSICAL THEORY OF INFLATION

Two Ways to View the Price Level

1. Price of Goods and Services

- ❖ The price level measures the cost of a representative basket of goods and services.
- ❖ A higher price level means consumers pay **more** for goods and services.

2. Value (Purchasing Power) of Money

- ❖ The price level also reflects the value of money.
- ❖ When the price level rises:
 - ✓ More money is required to buy the same goods.
 - ✓ Alternatively, each dollar buys **fewer goods and services**.
 - ✓ The purchasing power of money declines.
- ❖ **Relationship:**

Price Level ↑ ⇒ **Value of Money** ↓

THE CLASSICAL THEORY OF INFLATION

Measuring the Value of Money

- Let, **P = Overall Price Level** (measured by CPI or GDP Deflator)
- Then,

$$\text{Value of Money} = \frac{1}{P}$$

- **P** = Dollars needed to buy a basket of goods.
- **1/P** = Quantity of goods and services that **\$1** can purchase.

Example: Ice-Cream Economy

Price of One Cone (P)	Value of \$1 (1/P)
\$2	1/2 cone
\$3	1/3 cone

Conclusion

- As the **overall price level (P)** increases, the **value of money (1/P)** decreases.

THE CLASSICAL THEORY OF INFLATION

Money Supply, Money Demand, and Monetary Equilibrium

- The **value of money** is determined by the **forces of supply and demand**.
- Just as supply and demand determine the **price of goods**, they also determine the **value of money**.

Money Supply

- The **Central Bank** controls the money supply.
- **Open-market operations:**
 - **Selling government bonds** → Receives Money → **Money supply decreases**
 - **Buying government bonds** → Pays out money → **Money supply increases**
- For simplicity, the quantity of money supplied is treated as a **policy variable controlled by the Bangladesh Bank (BB)**.

Money Demand

- Money demand reflects **how much wealth people want to hold in liquid form**.
- Influenced by factors such as:
 - Use of **credit cards**
 - Availability of **ATMs**
 - **Interest rates**
 - **Average price level** (most important)

THE CLASSICAL THEORY OF INFLATION

Money Supply, Money Demand, and Monetary Equilibrium

Why Does the Price Level Affect Money Demand?

- ❖ Money is the **medium of exchange** used to buy goods and services.
- ❖ As the **price level increases**:
 - Transactions require **more money**.
 - People demand **more money** to make purchases.
- ❖ Therefore:
 - **Higher Price Level (P)**
→ **Lower Value of Money** → **Higher Money Demand**

Long-Run Monetary Equilibrium

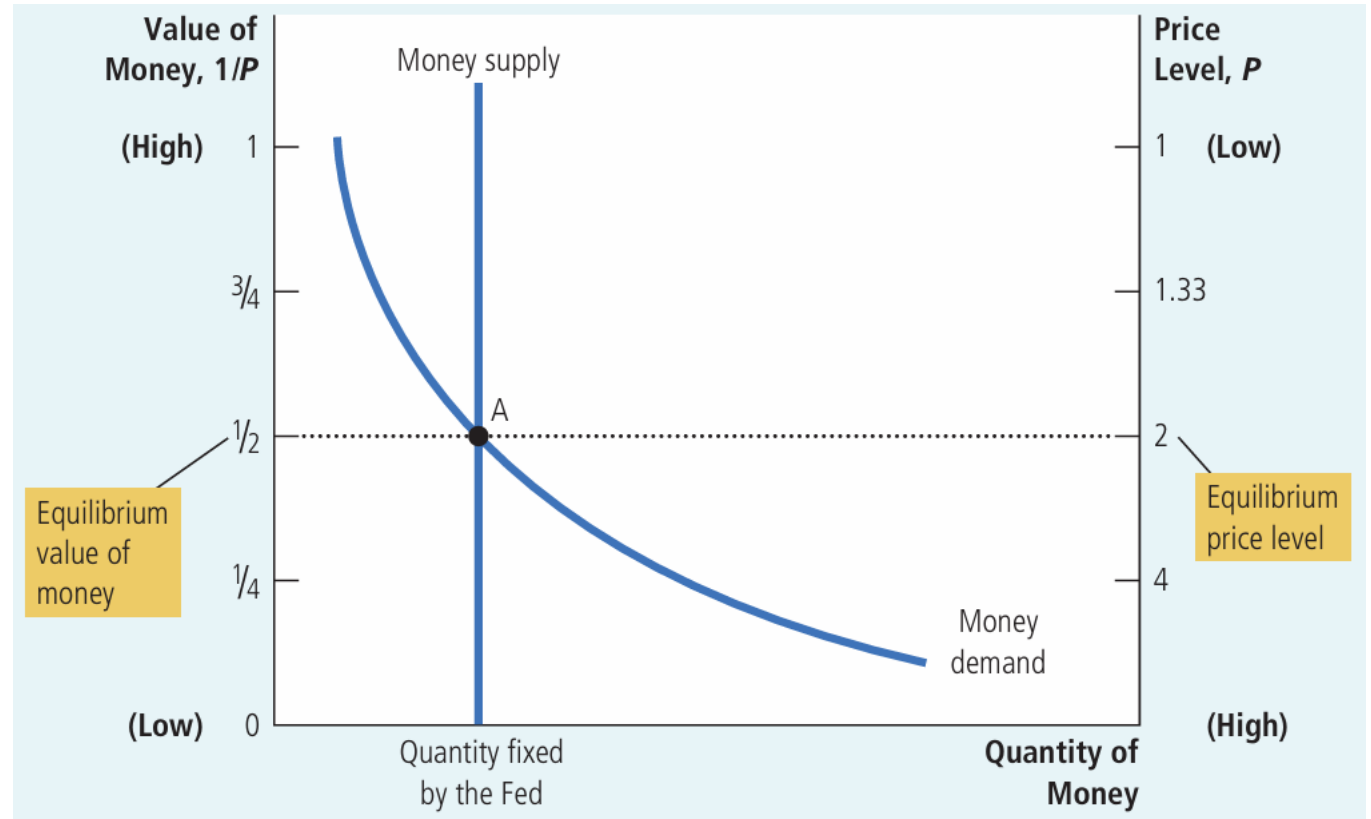
- ❑ In the long run, the **price level adjusts** to equate:
 - **Money Supply** (set by the **BB**)
 - **Money Demand** (by households and firms)
- ❑ If the **price level is too high**:
 - Money demand exceeds money supply.
 - The **price level falls** toward equilibrium.
- ❑ If the **price level is too low**:
 - Money supply exceeds money demand.
 - The **price level rises** toward equilibrium.
- ❑ **Equilibrium**: Money supplied = Money demanded.

THE CLASSICAL THEORY OF INFLATION

Money Supply, Money Demand, and Monetary Equilibrium

Money Market Equilibrium

- **Horizontal Axis:** Quantity of Money
- **Left Vertical Axis:** Value of Money ($1/P$)
- **Right Vertical Axis (Inverted):** Price Level (P)
- **Money Supply (MS) Curve:**
 - Vertical curve
 - Fixed by the **Central Bank**
- **Money Demand (MD) Curve:**
 - Downward sloping
 - When the **value of money falls** (P rises), people demand **more money**.



Equilibrium (Point A)

- **Money Supply = Money Demand**
- Determines:
 - **The value of money ($1/P$)**
 - **The equilibrium price level (P)**

THE CLASSICAL THEORY OF INFLATION

Effects of a Monetary Injection

What Is a Monetary Injection?

- A **monetary injection** occurs when the **Central Bank** increases the **money supply**.
- Example:
 - Printing additional money and distributing it.
 - More realistically, **buying government bonds through open-market operations**.

Effect on the Money Market

- The economy starts in **monetary equilibrium**.
- An increase in the money supply shifts the **Money Supply (MS)** curve **rightward**:
 - $MS_1 \rightarrow MS_2$
- Equilibrium changes from **Point A** to **Point B**.

THE CLASSICAL THEORY OF INFLATION

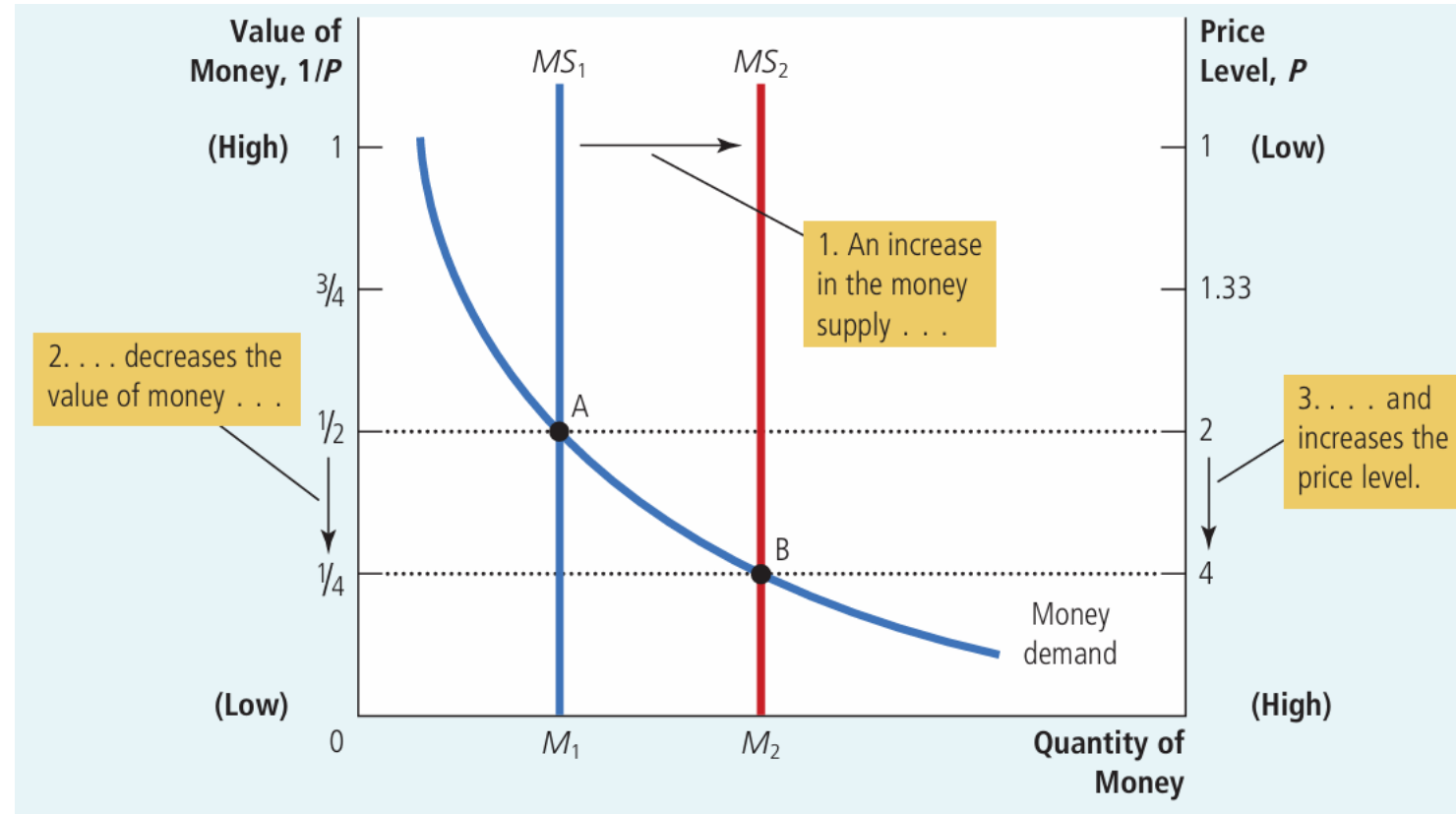
Effects of a Monetary Injection

Effects of Increased Money Supply

- **Value of money falls:** From $1/2$ to $1/4$
- **Price level rises:** From 2 to 4
- More money chasing the same quantity of goods leads to **higher prices**.
- Reduced purchasing power of money leading to inflation.

Quantity Theory of Money

- The **quantity of money** determines the **value of money**.
- Growth in the **money supply** is the **primary cause of inflation**.



Famous Statement

- Milton Friedman: **"Inflation is always and everywhere a monetary phenomenon."**

THE CLASSICAL THEORY OF INFLATION

Effects of a Monetary Injection: A Brief Look at the Adjustment Process

- A **monetary injection** initially creates an **excess supply of money** because people hold more money than they want at the current price level.
- People spend or lend their excess money by:
 - Buying **goods and services**, or
 - Purchasing **bonds** or depositing money in **banks**.
- This increases the **overall demand for goods and services**.

How the Economy Returns to Equilibrium

- The economy's **productive capacity** (labor, capital, natural resources, and technology) **does not change**.
- Higher demand raises the **overall price level**.
- As prices rise:
 - People need more money for transactions.
 - **Money demand increases**.
- A **new equilibrium** is reached when:
 - **Money Demand = Money Supply**.

Key Adjustment Process:

□ Monetary Injection → Excess Money → Higher Spending → Higher Prices → Higher Money Demand → New Monetary Equilibrium

THE COSTS OF INFLATION

An Essential *Economic Inquiry*

The Public Concern

Inflation is continuously monitored, scrutinized, and debated. It is widely perceived as a severe economic threat that directly affects the daily lives of citizens and the stability of global markets.

The Macro Challenge

But is this deep concern scientifically justified? If inflation is truly a problem, what specific mechanisms cause its damage? Economists focus on the real problems that price changes cause for businesses and the economy.

THE COSTS OF INFLATION

The *Inflation Fallacy*

A Fall in Purchasing Power?

Ask a typical worker why inflation is bad, and they will likely say it robs them of their hard-earned purchasing power.

The Buyer-Seller Paradox:

When prices rise, buyers naturally pay more. However, sellers of those goods and services simultaneously receive more.

Balanced Incomes:

Because most individuals earn income by selling their services and labor, nominal income inflation naturally goes hand-in-hand with price inflation.

THE COSTS OF INFLATION

The Principle of *Monetary Neutrality*



Psychological Bias

Workers interpret a nominal raise (e.g., 10%) as a direct reward for effort, feeling cheated when inflation (e.g., 6%) reduces the real value of their raise to only 4%.



Real Determinants

In reality, real incomes are determined by real macroeconomic variables: physical capital, human capital, natural resources, and available production technologies.



Nominal Adjustments

If the central bank dropped inflation from 6% to 0%, raises would simply fall from 10% to 4%. The worker might feel less cheated, but real income growth would remain identical.

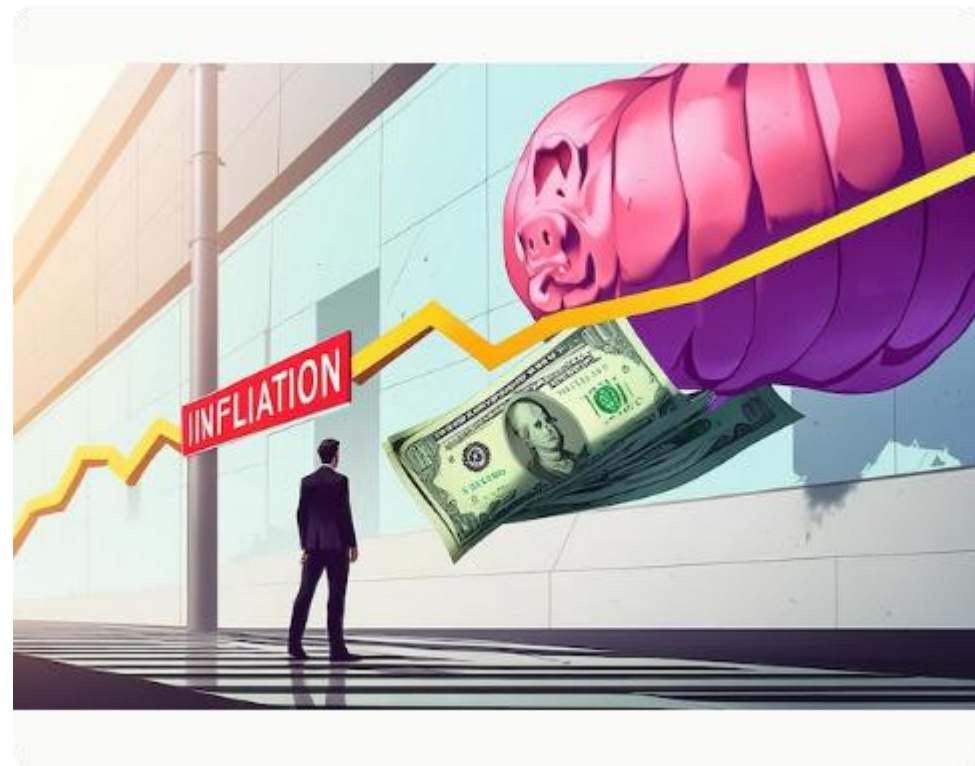
THE COSTS OF INFLATION

Why Is Inflation *A Problem?*

No Single Answer: If nominal incomes rise alongside prices, inflation's threat is not a simplistic loss of purchasing power, but rather a combination of multiple distinct economic frictions.

Adverse Effects on Real Variables: Economists have identified several distinct real costs of inflation, showing that persistent growth in the money supply does indeed disrupt the real economy.

Market Distortions: These structural costs create inefficiencies in transaction mechanisms, relative prices, tax systems, resource allocations, and business planning.



THE COSTS OF INFLATION: SHOELEATHER COST

Inflation as a Quiet Tax

The Inflation Tax

- ❖ Inflation acts as a silent tax on cash holders. Rather than taking physical notes directly, it quietly erodes the real purchasing power of the currency inside your wallet.
- ❖ This transfers real economic resources and buying power away from households and directly into the hands of the government.

The Deadweight Loss

- Like any traditional tax, the inflation tax distorts financial incentives. People alter their daily behavioral habits to avoid paying the tax altogether.
- This collective distortion causes deadweight loss as citizens waste scarce society resources to sidestep the erosion of their cash.

THE COSTS OF INFLATION: SHOELEATHER COST

How We Avoid the Tax

- ☐ To minimize losses from the continuous erosion of currency, individuals are forced to change how they hold their wealth:
- ☐ Reducing the overall quantity of physical cash held in wallets.
- ☐ Making more frequent, smaller trips to the bank to withdraw funds.
- ☐ Keeping maximum wealth stored in interest-bearing savings accounts.
- ☐ Trading cash quickly for stable physical goods or assets.



THE COSTS OF INFLATION: SHOELEATHER COST

The Real Definition

"The actual cost of reducing your money holdings is not the wear and tear on your shoes, but the time and convenience you must sacrifice to keep less money on hand than you would if there were no inflation."

— N. GREGORY MANKIW, PRINCIPLES OF ECONOMICS

THE COSTS OF INFLATION: SHOELEATHER COST

Bolivian Hyperinflation

Survival Rules

- ❖ During Bolivia's 1980s hyperinflation, prices soared at annual rates of 38,000%. Teacher Edgar Miranda had to immediately run to convert his monthly 25 million peso wage (\$50) into black-market U.S. dollars before the currency lost value by the hour.
- ❖ This desperate sprint represents the massive, exhausting magnitude of shoeleather costs in a runaway economy.



THE COSTS OF INFLATION: MENU COSTS

Understanding Menu Costs



Deciding & Planning

Firms must allocate critical executive time and research resources to decide on optimal new price points amidst unpredictable market shifts.



Administrative Expense

Firms must continuously print new catalogs, update retail stickers, reissue menus, and distribute updated pricing structures to distributors.



Customer Annoyance

Firms risk dealing with severe customer irritation and lost loyalty as rapid price adjustments upset standard consumer expectations.

THE COSTS OF INFLATION: MENU COSTS

Infrequent Price Adjustments

The Costly Transition

- Most businesses leave prices unchanged for weeks, months, or even years because updating them is inherently expensive. A typical U.S. firm only adjusts its pricing structure once a year.
- However, during periods of high inflation or hyperinflation, annual price schedules become entirely obsolete. Firms are forced to rewrite their prices daily, creating massive operational waste.



Sulṭān's Dine
سلطان داین
Dine In | Catering | Take Out

Take Away Only (Special Addition)		1/2	1
1. Multan Tehari (Take Away Only)		250	439
2. Multan Tehari			1/2
Take Away Only			
1. Kacchi (Basmali) Half			299
2. Kacchi (Basmali) Half with Barhani, Jarda/Firni			420
3. Kacchi (Basmali) Half with Chicken Roast, Barhani, Jali Kabab			549
Dine-In and Take Away Special			
1. Kacchi (Basmali) 1:2			799
Biryani			
1. Chicken Dum Biryani		199	580 899
Kacchi			
1. Kacchi (Basmali)		499	1349 2349
2. Kacchi (Basmali), Barhani, Jarda/Firni		620	1699 2999
3. Kacchi (Basmali), Chicken Roast, Barhani		699	1999 3349
Marag Palaw			
1. Marag Palaw		299	830 1340
2. Plain Palaw, Chicken Roast, Barhani		299	799 1350
Wedding Buzanza			
1. Kacchi Pfalter Kacchi (Basmali), Chicken Roast, Jali Kabab, Beef Rezala, Barhani, Jarda/Firni, Chutney, Salad		999	2449 4449
2. Plain Palaw Pfalter Plain Palaw, Chicken Roast, Beef Rezala, Jali Kabab, Barhani, Jarda/Firni, Chutney, Salad		649	1899 3099
Add Ons			
1. Plain Palaw	120		
2. Chicken Roast	150		
3. Jali Kabab	60		
4. Beef Rezala	200		
5. Chutney	25		
Beverage			
1. Water (Small/ Large)			M.R.P.
2. Soft Drinks (Pet Bottle)			M.R.P.
3. Barhani (1 Glass)			70
4. Zafarani Sorbet (1 Glass)			90
5. Laitan (500 ml)			170
6. Laitan (1 Liter)			340
7. Laitan (1 Glass)/(500 ml)			70/140
Desserts			
1. Firni	70		
2. Jarda	70		
Specials			
1. Whole Chicken Roast			599
2. Asta Khashi (Pre-Order)			18000